4FLOW
Your supply chain solution

AI in warehousing

Improving performance across the intralogistics lifecycle

Executive summary: the operational impact of AI in warehousing

Artificial intelligence (AI) is increasingly transforming modern warehouse and supply chain operations. While traditional technologies such as warehouse management systems (WMS), transportation management systems (TMS) and barcode scanning infrastructure have long supported logistics execution, AI introduces the ability for systems to analyze operational data and adapt decisions based on changing conditions.

Growing supply chain complexity is driven by several structural developments including the expansion of e-commerce fulfillment, increasing SKU assortments and rising expectations for delivery speed and service reliability. These conditions often expose the limitations of static rule-based warehouse processes.

AI technologies enable more adaptive and data-driven logistics operations. Machine learning models, computer vision systems and intelligent data processing platforms can improve operational visibility and optimize workflows across multiple warehouse processes.

This whitepaper examines how AI can be applied across the warehouse lifecycle including:

- yard management
- receiving operations
- warehouse execution
- inventory visibility
- outbound logistics

Each operational area is analyzed using a consistent framework that evaluates operational context, AI capabilities, technology architecture and representative vendors.

The objective of this paper is to provide a structured perspective on where AI can deliver practical operational improvements and how organizations can evaluate potential implementation strategies.

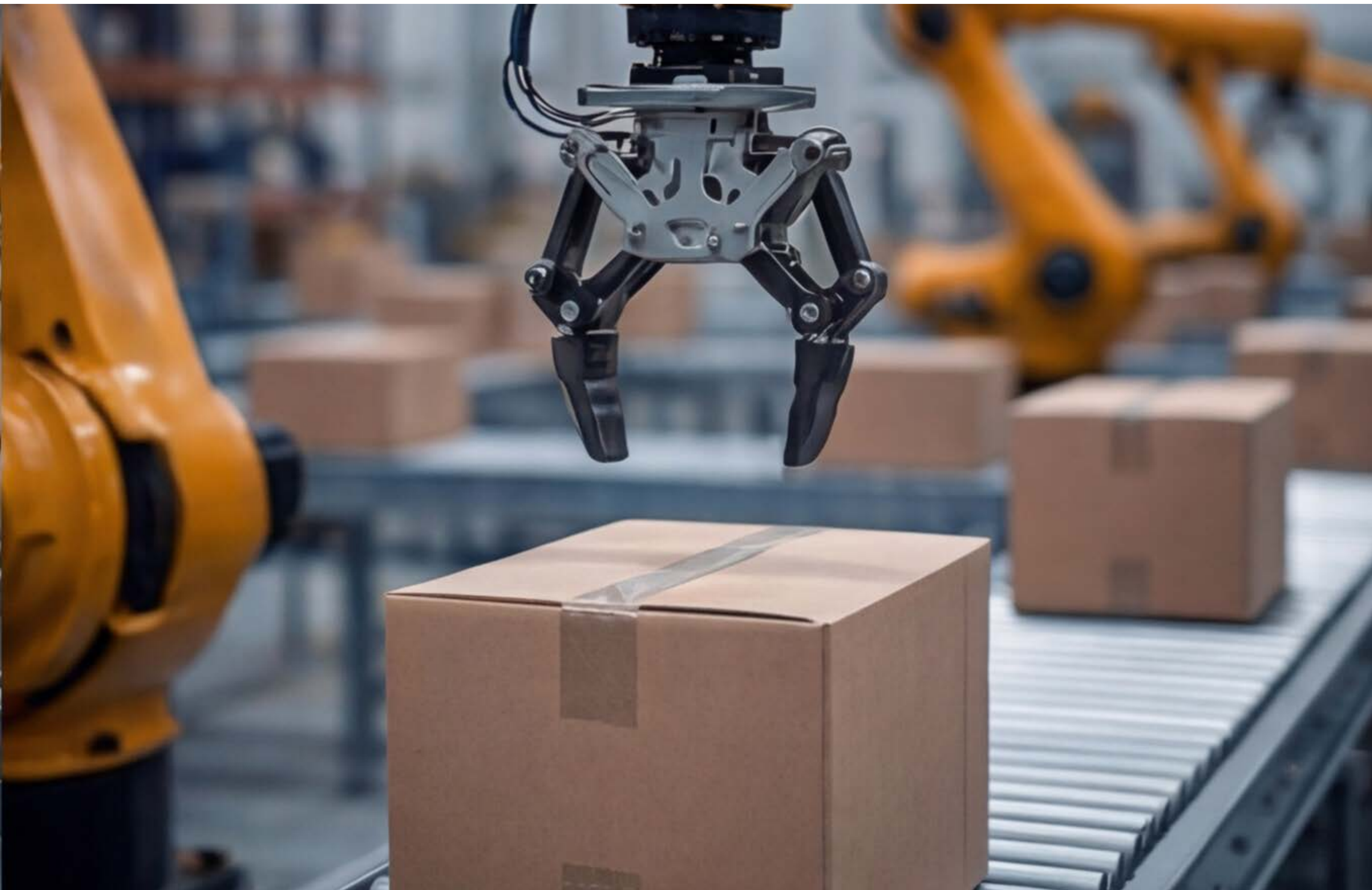
AI in intralogistics: why the shift is accelerating

AI has become an important technology influencing supply chain management and warehouse operations. Advances in machine learning, computer vision and natural language processing have expanded the practical applications of AI within logistics environments.

Warehouse operations generate large volumes of operational data through enterprise platforms such as WMS, TMS and enterprise resource planning (ERP) systems. AI technologies can analyze this data to support improved operational planning and workflow optimization.
Traditional warehouse systems typically rely on predefined operational rules for processes such as inventory storage, order picking and replenishment. While these systems improve efficiency, they often lack the ability to dynamically adapt to fluctuating order volumes, changing SKU demand patterns and varying labor availability.

AI provides the capability to analyze these dynamic operational conditions and generate insights that support more adaptive decision making across warehouse processes.

This paper examines how AI technologies are being applied across intralogistics operations and highlights representative technology vendors developing solutions within this evolving logistics technology landscape.



Analytical framework for evaluation: how to assess AI opportunities in warehousing

To evaluate AI applications within warehouse environments, this paper applies a consistent analytical framework across each operational stage of the warehouse process flow. This framework ensures that each section evaluates technological capabilities using comparable criteria. The analysis focuses on five dimensions.

Operational context

The operational role of each process within the warehouse lifecycle including inbound transportation coordination, receiving dock processing, inventory storage and outbound fulfillment across distribution centers (DCs). It also identifies common operational challenges encountered by organizations including dock congestion, inefficient pick paths and limited inventory visibility.

Role of AI

How AI can address the operational inefficiencies identified in the operational context. The analysis focuses on practical capabilities rather than theoretical technology potential including improvements to dock utilization, picking efficiency and inventory accuracy across warehouse operations.

Technology architecture

The technical systems enabling AI functionality. Examples include machine learning models, computer vision systems, data integration platforms and sensor networks deployed across warehouse facilities. Understanding the technology architecture helps organizations evaluate implementation requirements including infrastructure investments, system integration complexity and data pipeline architecture between enterprise systems such as WMS, TMS and ERP platforms.

Vendor landscape

Representative vendors are highlighted to illustrate how these technologies are currently being implemented in operational environments. The vendors included in this report represent organizations actively developing AI capabilities within intralogistics systems including yard management, warehouse orchestration and fulfillment automation platforms.

Strategic fit

Evaluation of which types of warehouse environments and operational contexts are most likely to benefit from each vendor's solution including high throughput distribution centers, e-commerce fulfillment facilities and complex multi-node logistics networks. Strategic fit analysis helps organizations understand where specific technologies provide the greatest operational value.

Applying this framework consistently across each operational stage provides a structured view of **how AI can improve warehouse performance** and supply chain efficiency.

AI in yard management: optimizing flow and visibility

Yard operations represent the interface between transportation networks and warehouse facilities. Trucks arriving at DCs must be checked in, assigned to dock doors and coordinated with internal warehouse operations including unloading schedules and labor allocation. Although yard management plays a critical role in overall logistics performance, it has historically received less technological attention than internal warehouse processes.

Many facilities still rely on manual coordination practices that include:

- radio communication between drivers and yard personnel
- manual entry of arrival information at gate checkpoints
- visual inspection of trailer identification numbers
- paper-based dock assignment procedures

These manual processes often result in limited real-time visibility into yard activity including trailer location, dwell time and queue length at facility gates.

When yard operations are not effectively coordinated several operational issues may occur:

- extended truck waiting times at facility gates
- inefficient dock utilization
- congestion within yard areas
- delayed inbound processing and dock-to-stock cycles

These inefficiencies can ultimately affect overall warehouse productivity and transportation costs. AI technologies are increasingly being applied to improve yard visibility and coordination across inbound transportation flows and logistics networks.

Role of AI in the process

AI improves yard management by providing real-time monitoring, predictive scheduling and automated coordination of trailer movements across the yard. AI-enabled yard management platforms integrate data from camera systems, TMS platforms and WMS platforms to maintain a continuously updated operational view of yard activity including trailer dwell times, estimated arrival times (ETA) and dock door availability.

These systems support several operational capabilities:

- automated identification of arriving trucks
- real-time monitoring of trailer locations across yard areas
- predictive analysis of dock scheduling requirements
- automated coordination between transportation arrivals and warehouse processing capacity

In advanced implementations AI models can generate predictive yard congestion forecasts and recommend proactive dock allocation strategies that reduce queue times and improve yard throughput.

By improving visibility and coordination, AI-driven yard management solutions help organizations reduce congestion, minimize detention costs and improve facility throughput across DCs.

How the technology works

AI-enabled yard management systems combine multiple technologies that allow continuous monitoring and analysis of yard activity.

Computer vision monitoring

Camera networks installed at facility entrances capture images of arriving vehicles. Computer vision models analyze these images to identify license plates and trailer numbers. These systems allow facilities to automatically register arriving trucks without manual data entry and support automated gate-in and gate-out processes.

Operational data integration

Information captured by computer vision systems is integrated with TMS and WMS platforms. This integration allows organizations to maintain a centralized operational view of inbound transportation activity including ETA, dock scheduling requirements and trailer status.

Predictive scheduling

Machine learning algorithms analyze historical operational data including inbound shipment patterns, advanced shipping notices (ASN) and dock processing times to forecast periods of high yard congestion and recommend optimized dock assignments. Predictive scheduling improves coordination between transportation arrivals and warehouse processing capacity.

Vendor examples

gatego

Gatego provides automated gate management systems that streamline the check-in process for truck drivers. Drivers can complete check-in procedures using QR codes while camera systems verify vehicle identification and trailer numbers.

EAIGLE

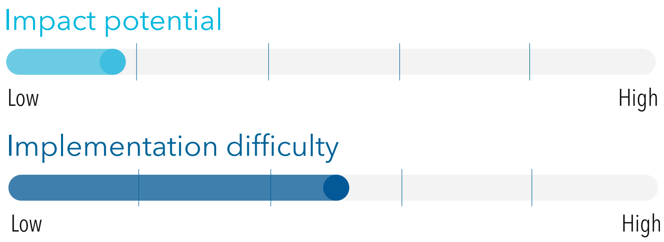
EAIGLE develops AI-powered camera platforms designed to monitor yard activity. These systems provide real-time visibility into trailer locations and operational activity across yard areas including dwell time and trailer movement patterns.

AUTO SCHEDULER

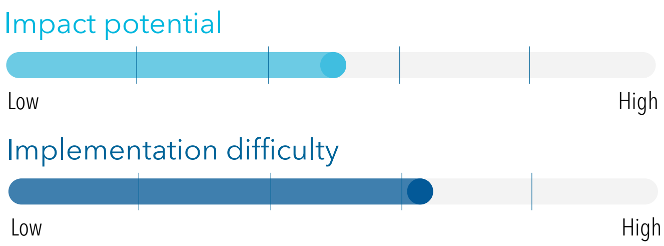
AutoScheduler.ai provides predictive scheduling software that analyzes operational data to coordinate dock assignments and trailer movements across distribution facilities.

Strategic fit by vendor

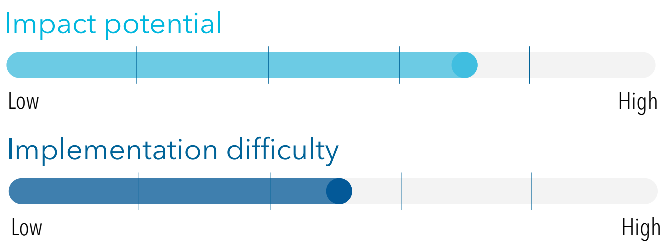
Gatego solutions are particularly effective for organizations seeking to automate gate check-in processes and reduce administrative workload at facility entrances. Physical security presence requirements will mitigate any fully automated solutions. Operations with physical security requirements will not gain fully automated potential from Gatego. Typical use cases include DCs experiencing high volumes of inbound and outbound truck traffic where gate congestion frequently occurs.



EAIGLE platforms provide continuous monitoring capabilities that are valuable for facilities requiring enhanced visibility into yard activity. The solution is particularly relevant for large DCs where manual monitoring of trailer locations and yard jockey activity is difficult.

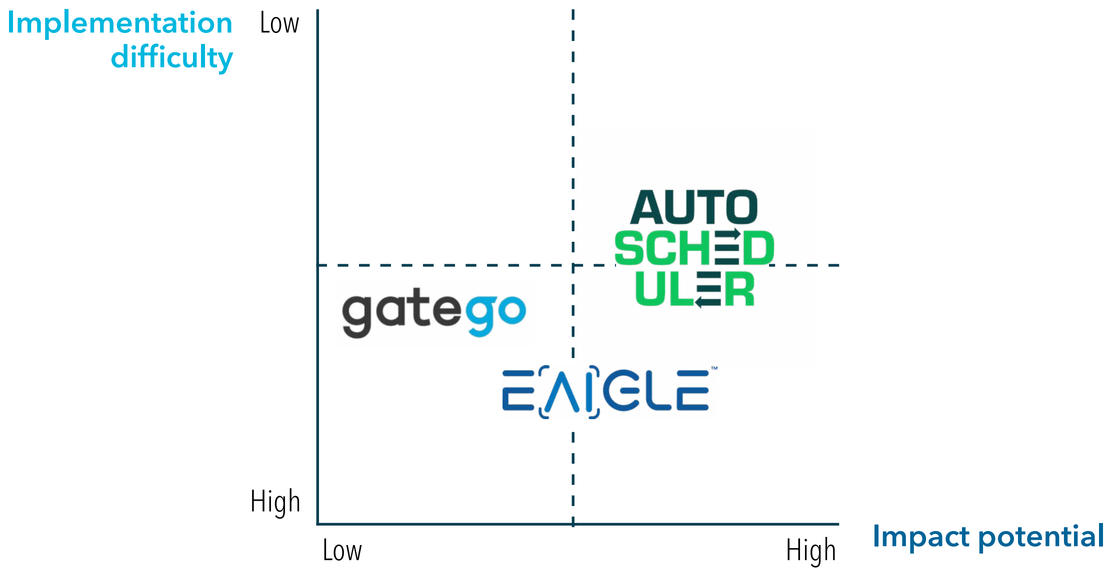


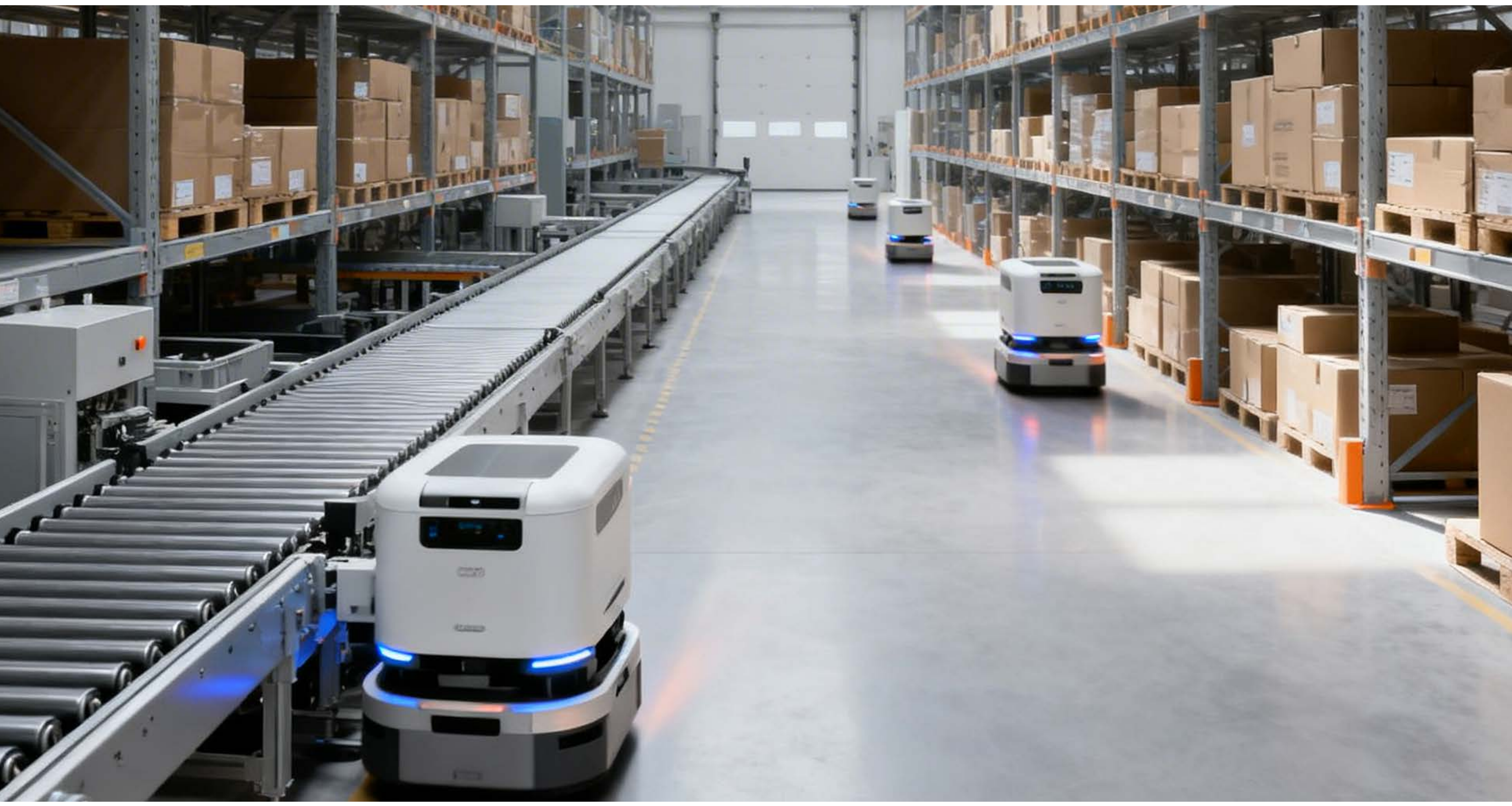
AutoScheduler.ai is best suited for complex distribution operations where dock scheduling coordination is a major operational challenge. Facilities with high transportation throughput and frequent dock congestion can benefit significantly from predictive scheduling capabilities and improved dock utilization.



The bottom line

AI improves yard management by replacing manual coordination processes with automated monitoring and predictive scheduling capabilities. These technologies improve operational visibility across the yard and enable more efficient coordination between transportation networks and warehouse operations.





AI in receiving operations: accelerating inbound accuracy and throughput

Receiving operations represent the first internal processing stage within warehouse environments. Once trucks have arrived at dock doors and shipments are unloaded, warehouse personnel must verify shipment contents, record inventory information and ensure that goods are accurately entered into WMS platforms.

Receiving processes typically include several operational activities:

- verification of shipping documentation such as bills of lading (BOL)
- validation of purchase order and ASN information
- inspection of inbound goods for damage or discrepancies
- recording of inventory quantities and units of measure (UOM)
- labeling and identification of individual items, pallets or cartons

Although these activities are fundamental to maintaining accurate inventory records, many receiving operations still rely heavily on manual processes. Workers often review paper documentation, manually enter information into ERP systems and visually inspect shipments to identify potential issues.

These manual processes introduce several operational challenges. Administrative data entry tasks can slow the receiving process and create bottlenecks during periods of high inbound volume at DCs. Human inspection processes may also produce inconsistent results depending on worker experience, process adherence, environmental conditions and workload levels at the receiving dock.

As DCs continue to process increasing shipment volumes, organizations are seeking new approaches to streamline receiving operations and reduce administrative workload.

AI technologies are increasingly being applied to automate several aspects of inbound processing including document interpretation, shipment verification and damage inspection. These technologies allow organizations to move toward digitally integrated inbound control processes, improving data accuracy and reducing dock-to-stock cycle time.

Role of AI in the process

AI improves receiving operations by automating tasks that previously required significant manual effort. AI technologies can analyze inbound documentation and automatically extract structured information that can be integrated directly into WMS and ERP platforms. This significantly reduces the time required to process inbound paperwork and improves synchronization between logistics documentation and physical inventory.

AI also improves consistency within inspection activities. Computer vision systems can analyze images of pallets, cartons and shipping units to identify potential damage, labeling errors or packaging issues.

These capabilities allow receiving operations to achieve several operational improvements:

- faster processing of inbound shipments
- reduced administrative workload for warehouse personnel
- improved consistency in inspection procedures
- faster integration of inbound inventory data into enterprise systems

In advanced implementations machine learning models can also detect discrepancies between shipment documentation, purchase orders and physical inventory received at the dock.

How the technology works

AI-driven receiving solutions typically combine several technologies that allow systems to process both structured and unstructured logistics information.

Intelligent document processing

Logistics documentation such as BOLs, packing lists and commercial invoices contain critical operational information. Intelligent document processing platforms use machine learning and optical character recognition (OCR) to extract key data fields including:

- shipment identifiers
- SKU information
- quantity and UOM
- supplier and carrier details

This information can then be automatically integrated into WMS or ERP platforms.

Computer vision inspection

Computer vision systems capture images of pallets and cartons during unloading processes. Machine learning models analyze these images to detect:

- damaged packaging
- missing labels
- incorrect pallet configurations

These systems enable automated inspection workflows that reduce manual quality control activities at receiving docks.

System integration

AI-generated data is integrated into enterprise logistics systems such as WMS, ERP and TMS platforms. Integration allows inbound inventory records to be updated automatically, improving inventory accuracy and operational visibility.

Vendor examples

ROSSUM

Rossum develops AI platforms designed to automate document processing tasks. The system uses machine learning models to interpret logistics documents and extract structured information from unstructured content such as BOLs, purchase orders and packing lists.

ABBYY

ABBYY provides intelligent document automation technologies that combine OCR with machine learning algorithms. These systems can process complex logistics documentation and convert it into structured digital records that integrate with ERP and WMS platforms.

ACI

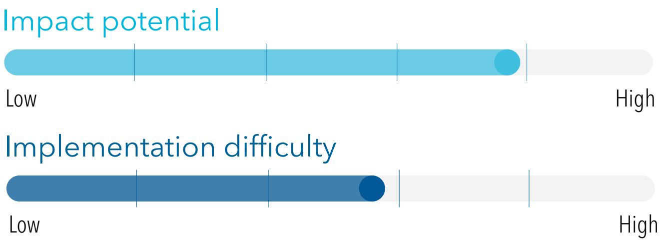
Automated Cargo Inspection develops computer vision systems designed to automate inspection processes for inbound shipments. The technology captures images of cargo and analyzes them to detect damage, packaging issues and dimensional inconsistencies.

KARGO

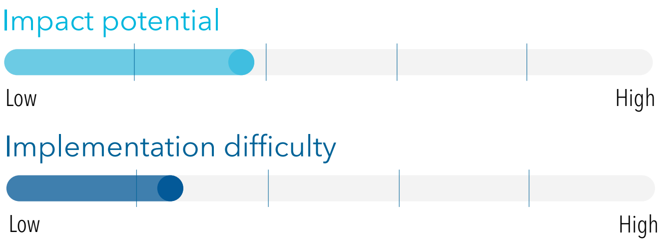
Kargo develops camera-based inspection towers installed at warehouse dock doors. These systems automatically capture images of pallets, cartons and handling units as they move through receiving operations.

Strategic fit by vendor

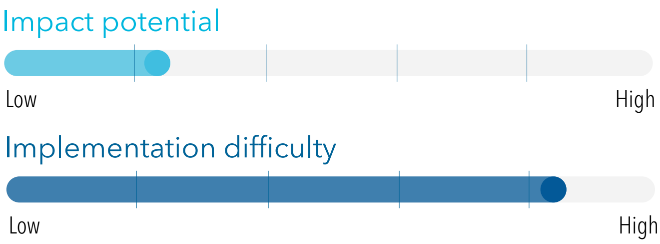
Rossum solutions are particularly well suited for organizations processing large volumes of logistics documentation. DCs that handle complex supplier documentation and high shipment volumes can benefit significantly from automated document interpretation.



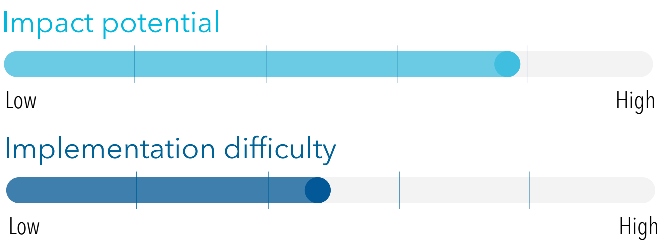
ABBYY technologies are most valuable in environments where document accuracy and regulatory compliance are critical. The platform is often used in operations requiring detailed validation of shipping documentation.



Automated Cargo Inspection systems are well suited for facilities seeking to improve the consistency of inbound inspection procedures. This technology is particularly useful for organizations handling fragile or high-value goods.

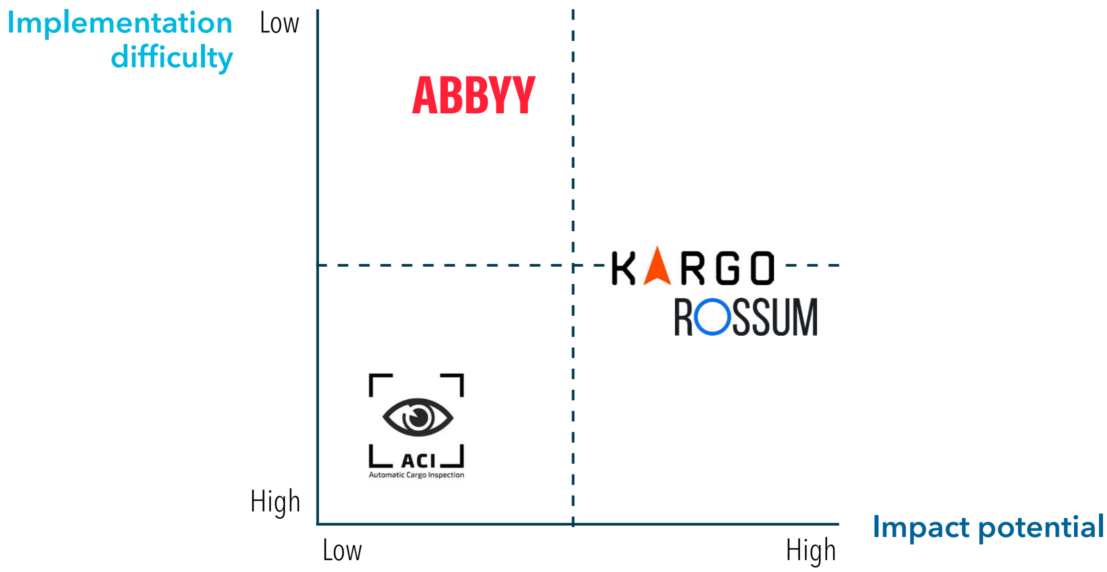


Kargo inspection towers provide strong value for DCs with high inbound throughput where manual inspection processes slow dock operations and delay dock-to-stock cycle times.



The bottom line

AI improves receiving operations by automating document interpretation and inspection activities. These technologies reduce administrative workload, improve processing speed and increase the accuracy of inbound inventory data. Over time these capabilities allow organizations to create digitally synchronized inbound logistics processes that improve receiving efficiency and supply chain visibility.



AI in warehouse execution: enhancing productivity through dynamic optimization

Once inventory has been received and recorded within WMS platforms, goods are transported to storage locations within the facility. At this stage operations transition to the core activities of warehouse execution.

Warehouse execution includes several operational processes:

- inventory storage and slotting
- order picking across discrete wave or batch workflows
- internal inventory movement using forklifts or automated equipment
- worker task coordination across warehouse zones

These activities represent the most labor-intensive portion of warehouse operations and account for a significant share of DC operating costs.

Traditional WMS platforms assign tasks using rule-based logic that determines how workers perform picking, replenishment and transport activities using handheld RF devices or voice-directed systems. However, these rules are often unable to adapt to dynamic operational conditions such as fluctuating order volumes or workforce availability.

AI technologies allow warehouse systems to continuously analyze operational data and dynamically optimize workflows within DCs.

Role of AI in the process

AI improves warehouse execution by enabling adaptive optimization of operational workflows. Machine learning models analyze operational data to optimize:

- inventory slotting strategies
- worker travel paths and pick routes
- order picking sequences
- coordination between human workers and robotic systems

These capabilities reduce travel distance, improve picking productivity and increase overall warehouse throughput.

AI systems can also dynamically assign tasks to workers based on real-time conditions such as workload distribution, worker location and order priority from the order management system (OMS).

How the technology works

AI-driven warehouse execution systems analyze data generated by WMS platforms, RF devices and automation platforms.

Machine learning optimization

Machine learning algorithms analyze historical operational data to identify patterns in order demand, SKU velocity and picking frequency. These insights allow systems to recommend optimized slotting strategies and task sequencing.

Worker guidance systems

AI-driven task management systems provide workers with real-time instructions through RF devices, voice picking systems or wearable technologies. These systems dynamically adjust task assignments based on operational conditions.

Robotic coordination

AI platforms can also coordinate activities between human workers and robotic systems such as autonomous mobile robots (AMRs) and automated guided vehicles (AGVs). These systems allocate tasks between humans and robots to maximize operational efficiency.

Vendor examples

GreyOrange

GreyOrange develops robotic automation platforms and warehouse orchestration software designed to coordinate warehouse operations across humans, robots and inventory systems.

Lucas

Lucas Systems provides AI-powered warehouse optimization tools including the Jennifer AI platform which combines voice-directed picking with machine learning optimization.

BRIGHTPICK

Brightpick develops autonomous robotic systems capable of retrieving inventory from storage locations and performing order picking activities.

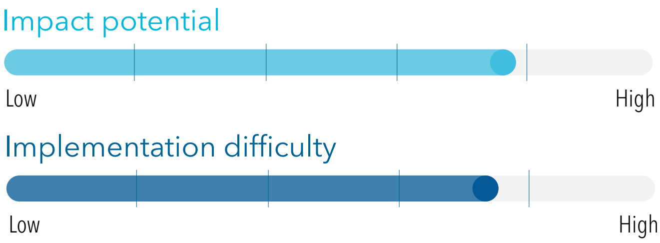
MAGAZINO

Magazino develops intelligent mobile robots designed for warehouse environments. These robots are coordinated through the proprietary AI platform.

Strategic fit by vendor

GreyOrange – GreyMatter Software

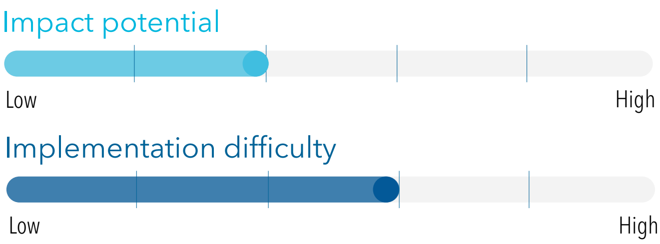
GreyOrange platforms are particularly effective for large fulfillment operations implementing integrated robotics systems across multiple warehouse processes.



GreyOrange

Lucas Systems – Jennifer AI Software

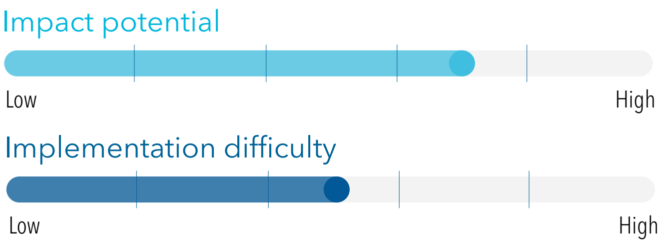
Lucas Systems technologies are well suited for warehouses seeking to improve worker productivity without implementing extensive robotic automation.



Lucas

Brightpick - Autopicker

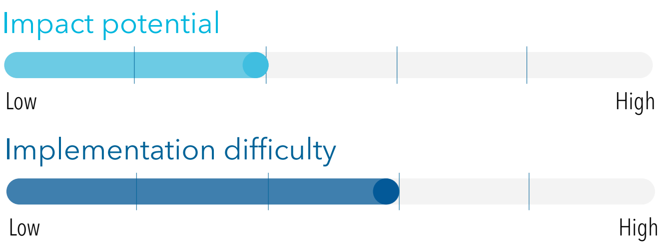
Brightpick solutions are especially relevant for e-commerce fulfillment environments with high order volumes and complex SKU assortments.



BRIGHTPICK

Magazino – SOTO and TORU

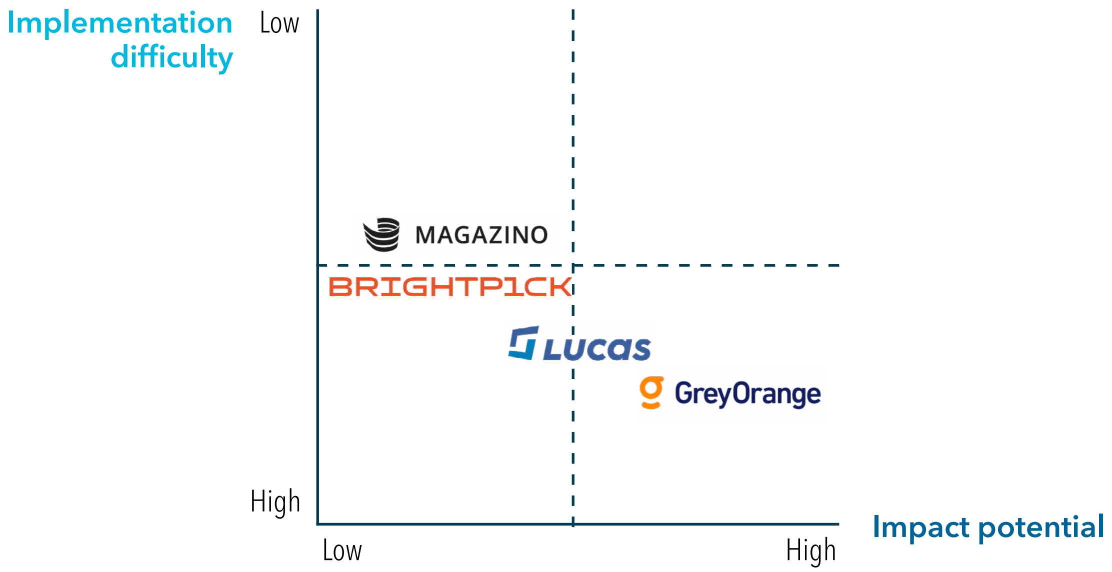
Magazino robotic platforms are effective in warehouse environments requiring flexible automation that can operate alongside human workers. Selecting the right platform requires careful evaluation of automation maturity, labor structure and warehouse throughput requirements.



MAGAZINO

The bottom line

AI improves warehouse execution by enabling dynamic optimization of operational workflows. These systems improve labor productivity, reduce travel time during picking operations and support coordination between human workers and robotic systems.



AI in inventory visibility: enabling continuous accuracy across storage locations

Accurate inventory visibility is essential for effective warehouse operations. WMS platforms rely on accurate inventory records to coordinate storage, picking and replenishment processes.

When inventory records become inaccurate warehouses may experience several operational issues including:

- increased time spent searching for misplaced inventory
- delays in order fulfillment and shipment processing
- increased labor requirements for manual cycle counting
- incorrect shipments to customers

Many facilities rely on periodic cycle counting procedures to maintain inventory accuracy. However, cycle counting can require significant labor resources and may disrupt normal warehouse operations.

AI technologies are increasingly being used to automate inventory monitoring and improve data accuracy across warehouse facilities.

Role of AI in the process

AI improves inventory visibility by continuously monitoring inventory locations and identifying discrepancies between physical inventory and system records. AI-driven inventory monitoring allows warehouses to move away from periodic cycle counting toward continuous inventory verification.

Operational benefits include:

- automated monitoring of inventory across large warehouse facilities
- faster identification of inventory discrepancies
- reduced labor requirements for manual cycle counting
- improved accuracy of WMS records

How the technology works

AI inventory visibility systems combine computer vision technologies with autonomous scanning platforms.

Autonomous scanning systems

Some solutions deploy drones or mobile robots that travel through warehouse aisles scanning storage locations and capturing images of barcodes or product labels.

Computer vision analysis

Machine learning models analyze captured images to identify SKU information and verify storage locations.

System synchronization

Extracted inventory information is synchronized with WMS platforms allowing discrepancies to be detected and corrected.

Vendor examples

GATHER AI

Gather AI develops autonomous drone systems designed to scan warehouse inventory and verify storage locations. Gather AI drones are capable of conducting cycle counts in cold storage environments. Gather AI[18]

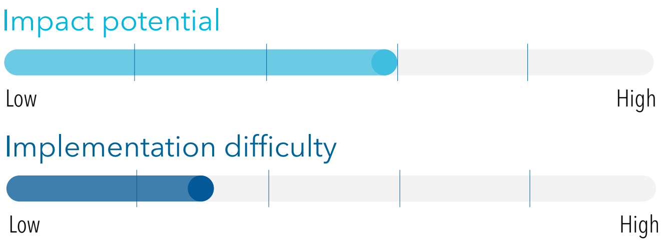
SCANDIT

Scandit provides advanced barcode scanning software that improves mobile data capture capabilities for warehouse workers. Scandit[19]

Strategic fit by vendor

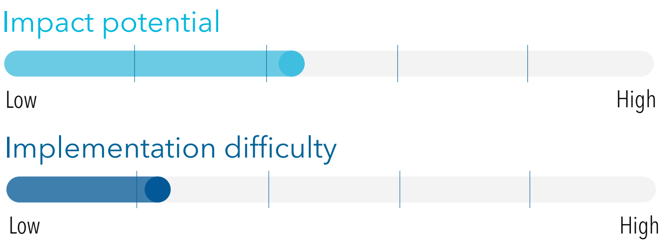
Gather AI

Gather AI systems are most effective for large DCs where manual cycle counting is time consuming and difficult to execute consistently.



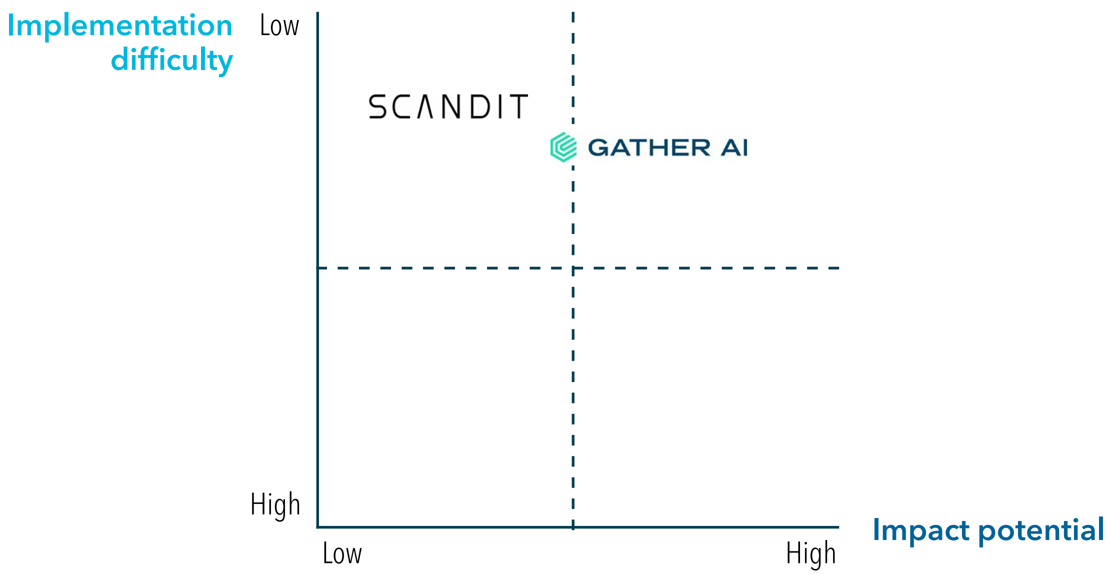
Scandit

Scandit technologies are best suited for operations seeking to improve barcode scanning efficiency without deploying additional hardware infrastructure.



The bottom line

Artificial intelligence improves inventory visibility by automating monitoring processes and AI improves inventory visibility by automating monitoring processes and reducing reliance on manual cycle counting activities. These systems enable warehouses to maintain near real-time inventory accuracy, improving both operational performance and customer service levels.



AI in outbound logistics: improving packing, sorting and shipment readiness

Outbound logistics represents the final stage of warehouse operations. After customer orders have been picked, warehouses must prepare shipments for transportation and final delivery.

Outbound activities include:

- packing individual products into cartons
- selecting optimal carton sizes based on dimensional weight (DIM weight) considerations
- palletization of shipments
- parcel sorting and carrier routing

These processes ensure that customer orders are correctly assembled and prepared for transportation across parcel networks and freight carriers.

Role of AI in the process
AI improves outbound logistics by optimizing packaging decisions and enabling robotic parcel handling systems. Machine learning models can analyze order composition, product dimensions and carrier requirements to determine optimal packaging configurations. Computer vision systems allow robotic platforms to identify parcels and manipulate them during sorting and palletization processes.

How the technology works
AI-driven outbound systems combine machine learning optimization with robotics and computer vision.

Packaging optimization
Machine learning algorithms analyze product characteristics to determine optimal carton sizes, packing configurations and palletization strategies.

Robotic parcel handling
Computer vision systems identify parcels and determine appropriate gripping points for robotic manipulation.

Automated sortation
Robotic systems direct parcels to appropriate shipping lanes based on carrier routing rules and TMS instructions.

Vendor examples

Packsize develops packaging automation systems that determine the optimal carton size for each shipment. Packsize[20]

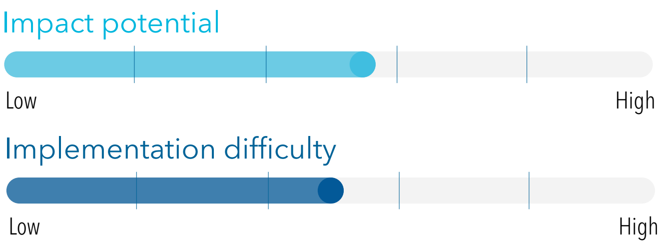
Ambi Robotics develops robotic parcel sorting systems that use machine learning and computer vision technologies. Ambi Robotics[21]

Berkshire Grey develops robotic automation systems designed to automate parcel handling and fulfillment operations. Berkshire Grey[22]

Strategic fit by vendor

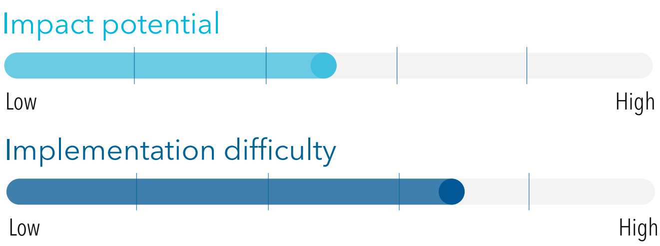
Packsize

Packsize systems are particularly effective for e-commerce fulfillment operations seeking to reduce packaging waste and shipping costs.



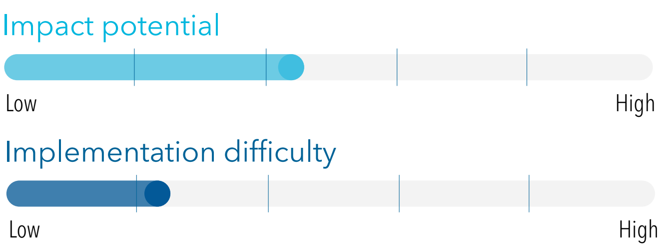
Ambi Robotics

Ambi Robotics solutions are well suited for high-volume parcel sorting environments.



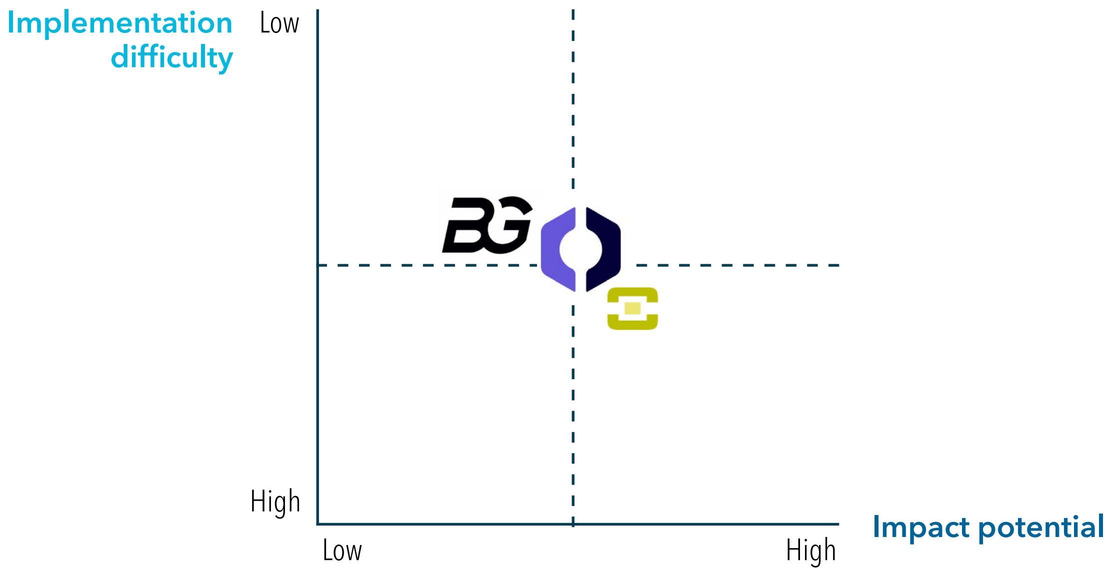
Berkshire Grey

Berkshire Grey systems are best suited for organizations implementing large-scale warehouse automation strategies.



The bottom line

AI improves outbound logistics by enabling more efficient packaging decisions and supporting robotic parcel handling operations. These capabilities support higher fulfillment throughput, improved order accuracy and more efficient DC operations.



Conclusion:
unlocking end-to-end value with AI across intralogistics

AI is becoming an essential component of modern warehouse operations as organizations manage increasing complexity across intralogistics networks. Across yard management, receiving, warehouse execution, inventory visibility and outbound logistics, AI provides practical capabilities that improve visibility, optimize resources and support data-driven decision making.

AI delivers the most value when integrated with existing systems such as WMS, TMS and ERP. Connecting real-time operational data with machine learning and computer vision reduces manual effort, increases accuracy and improves throughput across distribution centers.
From a 4flow perspective, effective AI adoption requires readiness across data quality, processes and system integration. Successful initiatives start with focused use cases that demonstrate measurable gains and then scale across the network. We work with organizations to evaluate how AI can be integrated into existing logistics operations and to define implementation paths that deliver sustainable performance improvements.

Organizations that treat AI as a core enabler within an integrated supply chain architecture will be best positioned to achieve lasting improvements across the warehouse lifecycle.

Authors

Florian Salamon
4flow consulting

Florian Salamon is a Principal at 4flow consulting with extensive experience in supply chain and logistics consulting, with a strong focus on intralogistics. He supports end-to-end supply chain transformations, including network and footprint design, operational optimization, automation and supply chain software implementations. With 4flow since 2012, Florian relocated to the United States in 2020, where he heads the Atlanta office and serves as a senior leader for intralogistics consulting as well as the automotive, industrial and aviation sectors across 4flow consulting North America. He holds a Master's degree from Heriot-Watt University in Edinburgh.

Martin Wilson
4flow consulting

Martin Wilson is a Manager at 4flow consulting with over four years of experience in supply chain and logistics consulting. He specializes in warehouse design, automation solutions, end-to-end supply chain transformations, network design and shopfloor and factory optimization. Martin has supported clients across the automotive, aviation, industrial, CPG and retail sectors. He holds a Master's degree from Georgia State University.

Santiago Gunther
4flow consulting

Santiago Gunther is a Consultant at 4flow consulting with experience in supply chain and logistics consulting, with a strong focus on intralogistics. He supports projects across warehouse planning and design, shopfloor and operational optimization and automation initiatives. Since joining 4flow in 2022, Santiago has contributed to projects within the automotive, industrial and aviation sectors, helping clients improve operational efficiency and implement sustainable logistics solutions.

Interested or have questions?
Contact us!

Get in touch to have your questions answered

4FLOW

North America

Detroit | 306 S Washington Ave | Ste 500 |
Royal Oak, MI 48067 | (313) 777-8300

Europe

Berlin | Hallerstrasse 1 |
10587 Berlin | Germany

4flow.com